

Anish Madan



Email : anishmad@cs.cmu.edu
madanish223@gmail.com

Webpage : anishmadan23.github.io

RESEARCH INTERESTS

- 3D Vision
- Foundational Models
- Digital Twins
- Vision for Robotics

EDUCATION

- **Carnegie Mellon University (CMU)** Pittsburgh, USA
Masters of Science, Robotics; CGPA: 4.09/4.00
Advisor: [Prof. Deva Ramanan](#) Aug 2022 - Aug 2024
- **Indraprastha Institute of Information Technology (IIIT), Delhi** India
B.Tech, Computer Science and Applied Maths; CGPA: 8.78/10.00
Advisor: [Prof. Saket Anand](#) Aug 2016 - Jan 2021

CONFERENCE PUBLICATIONS

- | | |
|---|--|
| Simultaneous Map and Object Reconstruction
<i>N. Chodosh*, A. Madan*, S. Lucey, D. Ramanan.</i> | 3DV 2025
Webpage , Paper |
| Revisiting Few-Shot Object Detection with Vision-Language Models
<i>A. Madan*, N. Peri*, S. Kong, D. Ramanan.</i> | NeurIPS D&B 2024
Paper , Code |
| SLI-pSp: Injecting Multi-Scale Spatial Layout in pSp
<i>AN Mathur*, A. Madan*, O. Sharma</i> | WACV 2023
Paper |
| REGroup: Rank-aggregating Ensemble of Generative Classifiers for Robust Predictions
<i>L. Tiwari, A. Madan, S. Anand, and S. Banerjee</i> | WACV 2022
Paper , Code |
| B-SMALL: A Bayesian Neural Network approach to Sparse MAML
<i>A. Madan, R. Prasad.</i> | ICASSP 2021
Paper , Code |
| Category Consistent Cyclic Visual Question Generation
<i>S. Uppal*, A. Madan*, S. Bhagat*, Y. Yu, R. Shah.</i> | MMAAsia 2020
Paper , Code |

WORKSHOP CHALLENGES AND PUBLICATIONS

- | | |
|---|--|
| Foundational FSOD Challenge
<i>A. Madan*, N. Peri*, S. Kong, D. Ramanan</i> | VPLOW @ CVPR 2024
Paper Challenge |
| NurtureNet:A Multi-task Video-based Approach for Newborn Anthropometry
<i>Y. Khandelwal, ..., A. Madan, ..., M. Tapaswi</i> | CVPM @ CVPR 2024
Paper |

* Equal Contribution

ACADEMIC EXPERIENCE

- **CMU Center for Autonomous Vehicle Research | CMU** Pittsburgh, USA
Graduate Research Assistant | Advisor: Prof. Deva Ramanan | NeurIPS '24, 3DV '25 Oct 2022 - Present
 1. **Multi-Modal Dynamic Reconstruction of Urban Scenes**
 - Leading efforts to extend SMORE to support multiple modalities (RGB and LiDAR)
 2. **Automating Object Detection Annotation Pipelines with Foundation Models**
 - Reformulated few-shot detection(FSOD) as alignment of foundational Vision-Language Models (VLMs) to multi-modal annotator instructions.
 - Proposed a new Foundational FSOD benchmark on nuImages.
 - Filed a patent based on this work (*currently under review*)
- **Graphics Research Group | IIIT-Delhi** New Delhi, India
Research Intern | Advisor: Dr. Ojaswa Sharma | WACV '23 Jan 2021 - July 2022
 - Proposed a StyleGAN based Image-to-Image Translation model to explicitly capture fine-grained semantics and structures in addition to the global style.
 - Improved FID performance by 16 points over SOTA on CelebA (Label-to-Face generation).

- **Bayesian View of Meta-Learning | IIIT-Delhi** New Delhi, India
Undergraduate Researcher / Advisor: *Dr. Ranjitha Prasad* / ICASSP '21 Jan 2020 - Jan 2021
 - Proposed a Bayesian neural network based approach to MAML to alleviate the problem of overfitting by introducing network sparsity.
- **Infosys Center of Artificial Intelligence | IIIT-Delhi** New Delhi, India
B.Tech Thesis / Advisor: *Dr. Saket Anand* / WACV '22 Aug 2018 - Feb 2020
 - Developed a novel adversarial defense technique to secure Deep Neural Networks against adversarial attacks.
 - Benchmarked on ImageNet and demonstrated performance at-par with SOTA whilst requiring only a fraction of the data.

INDUSTRY EXPERIENCE

- **Bosch Research** Pittsburgh, USA
Machine Learning Research Intern May 2024 - Aug 2024
 - Conducted large scale experiments for **dynamic scene reconstruction** for urban scenes using LiDAR
 - Achieved **10x** improvement in **LiDAR Novel View Synthesis** over SOTA
- **Wadhvani AI** New Delhi, India
Associate ML Scientist / Advisor: *Dr. Makarand Tapaswi* Jan 2021 - July 2022
 - **Newborn Anthropometry Project:** Developed 3D mesh reconstruction based (SMPL, HMR) and parametric model-free (Contrastive Learning, Transformers) techniques to analyse low-birth weight babies from a video.
 - Contributed in developing a dataset of baby keypoints and joints.
 - * Conducted annotation training exercises with external agencies.
- **Robert Bosch Research and Technology Centre** Bangalore, India
Research Intern Aug 2019 - Dec 2019
 - Developed context-based *cut-and-paste* data augmentation techniques for semantic segmentation.
 - Improved performance by 5 AP on various categories, e.g., pedestrians on Bosch's Autonomous Vehicle dataset.
- **IIIT-D Autonomous Last mile Vehicle (ALIVE)** New Delhi, India
Summer Intern / Advisor: *Dr. Saket Anand* May 2018 - July 2018
 - Developed methods to detect speed breakers for AVs and improved existing lane detection algorithms.

SERVICES

- **Reviewer:** CVPR 2025, ECCV 2024, ICLR 2023
- **Mentorship @ CMU:** CMU AI Mentoring Program 2023
- **Mentorship @ Wadhvani AI :** Guided 2 interns for their projects on Uncertainty Estimation, Imbalanced Regression
- **Challenge Organizer:** Visual Perception and Learning in an Open World (**CVPR 2024**)
- **Invited Talks:** *AI Winter School*, IIIT-Delhi, 2019
- **Volunteering @ CSSAR NGO:** Taught basics of Science and English to students of classes 3-10.

AWARDS AND ACHIEVEMENTS

- **Dean's Award for Academic Excellence:** Excellent academic performance in a calendar year 2019
- **Technical Paper Award:** National Rank 1 for technical paper at [AUVSI-SUAS](#) competition 2017
- **Joint Entrance Examination (JEE):** Among top 0.96% out of 1.2 million candidates 2016